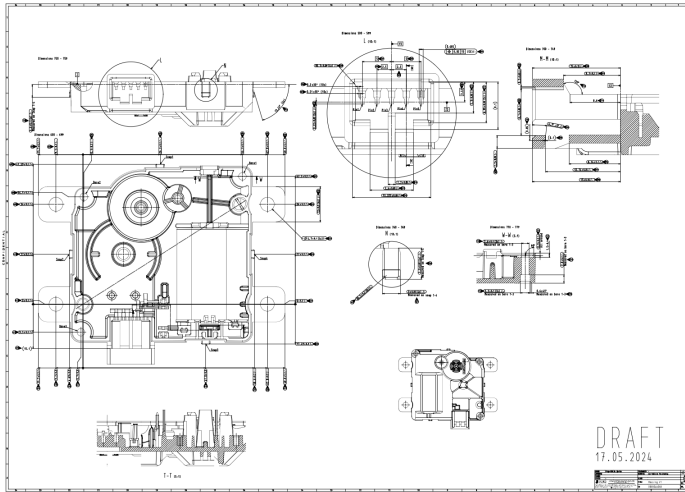


Algorithm

Given Sample 3D design Diagram.

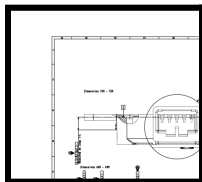


Divide diagram into tiles. Advantage is faster to predict and increase over all accuracy. We need to fix the dimension of the diagram, for all the images.

Training data set details:

Image ID: IMG_001

Tile ID: Tile_001



OBJECT/COMPONENT:

object_Bounding_Box : <x,y,width,height> #coco Format

classification_type: Object_id # Machine name/component Type Id

OCR:

content_bbox : <x,y,width,height>

text_extracted : Text #text within boundingbox

Annotation Arrow:

Points:[x1 y1 x2 y2]

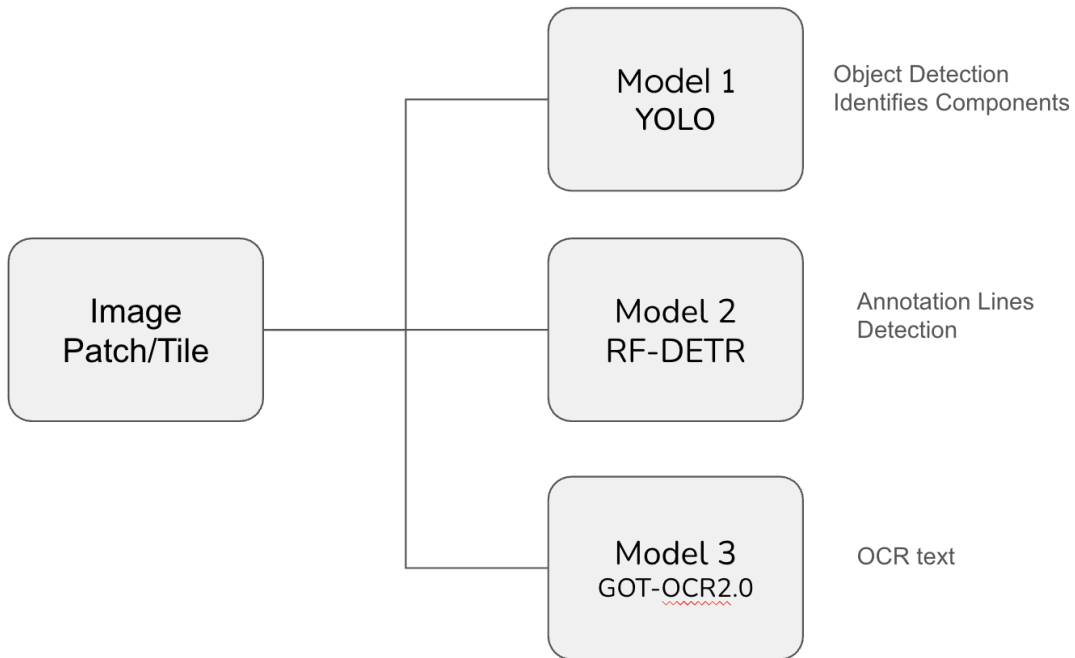
class_id: Arrow Type # annotation line type

Sample_Data:

```
{
  "Image ID": "IMG_001",
  "Tile ID": "Tile_001",
  "OBJECT/COMPONENT": [
    {
      "object_Bounding_Box": [120, 340, 85, 42],
      "classification_type": "Ctype_1"
    },
    {
      "object_Bounding_Box": [280, 150, 60, 60],
      "classification_type": "Ctype_2"
    },
    {
      "object_Bounding_Box": [410, 220, 30, 30],
      "classification_type": "Ctype_3"
    }
  ],
  "OCR": [
    {
      "content_bbox": [122, 342, 80, 38],
      "text_extracted": "Ø25.4"
    },
    {
      "content_bbox": [295, 165, 30, 30],
      "text_extracted": "07"
    }
  ],
  "Annotation Arrow": [
    {
      "Points": [205, 361, 280, 361],
      "class_id": "leader_line"
    },
    {
      "Points": [410, 235, 500, 235],
      "class_id": "section_arrow"
    }
  ]
}
```

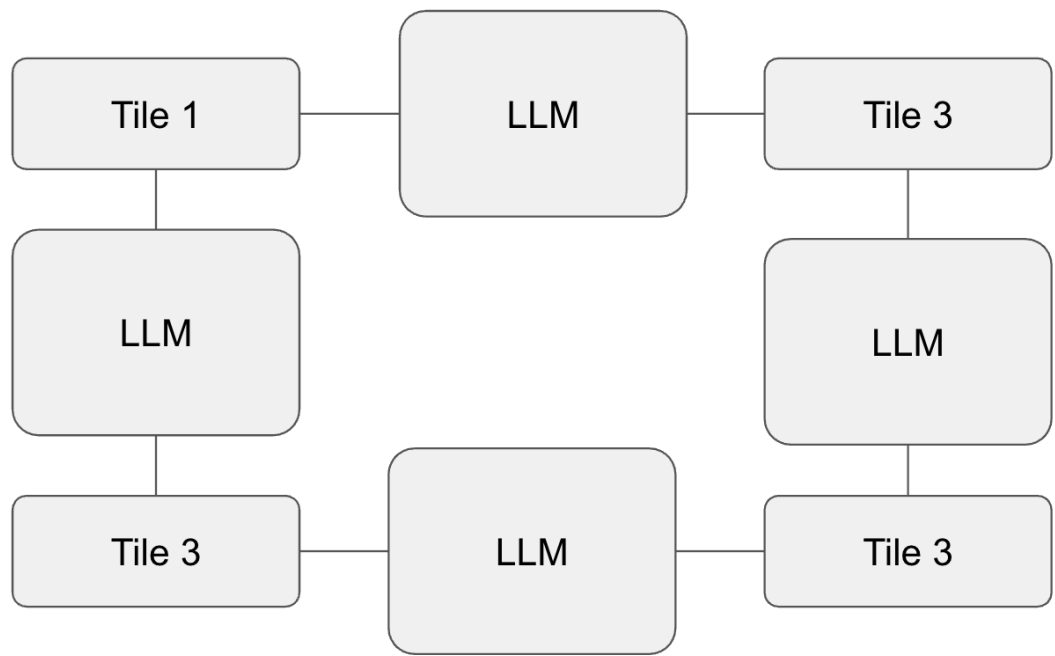
Stage 1: Meta extraction

We pass each tile pass, three models.



Each model will capture the BBox of components, Annotation lines to extract the meta data like dimensions, and OCR to extract all textual information.

Stage 2: Patch Network



Two tiles will be sent to LLM to merge the Points and rename the Class, annotation text. This will improve the accuracy.

Point Merging technique.

TILE 1

	Point, Class_id			
--	-----------------	--	--	--

TILE 2

	Point, Class_id			
--	-----------------	--	--	--

Merge Two points and Class ID based on the input from LLM.

Final JSON. will be combined merged JSON of each tile.